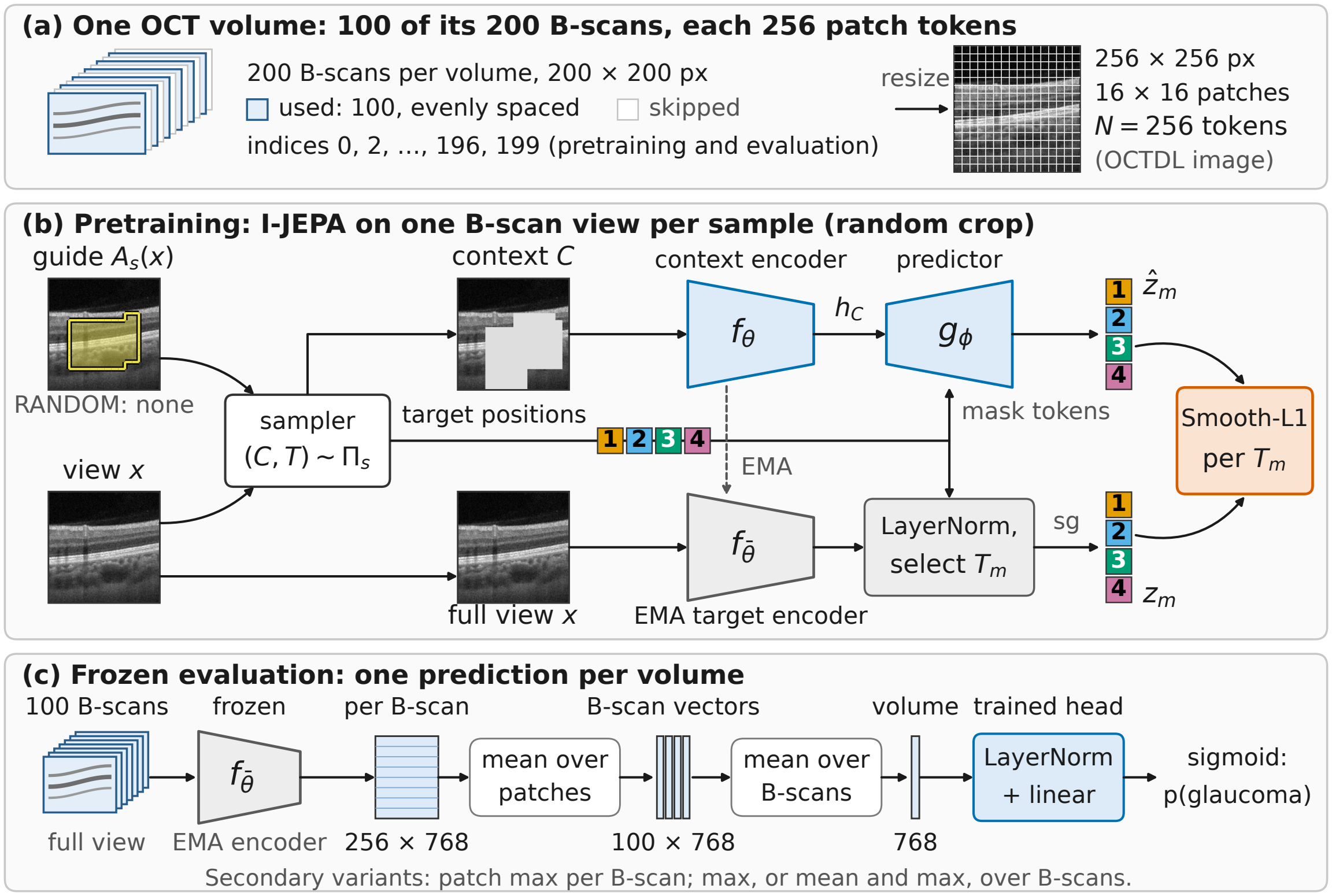

(a) One OCT volume: 100 of its 200 B-scans, each 256 patch tokens
200 B-scans per volume, 200 × 200 px
used: 100, evenly spaced
skipped
indices 0, 2, …, 196, 199 (pretraining and evaluation)
resize
256 × 256 px
16 × 16 patches
N = 256 tokens
(OCTDL image)
(b) Pretraining: I-JEPA on one B-scan view per sample (random crop)
guide $A_s(x)$
context C
context encoder
predictor
f_θ
h_C
g_ϕ
$\hat{z}_m$
RANDOM: none
sampler
$(C, T) \sim \Pi_s$
target positions
1 2 3 4
mask tokens
Smooth-L1
per T_m
view x
EMA
full view x
$f_{\bar{\theta}}$
LayerNorm,
select T_m
sg
z_m
EMA target encoder
(c) Frozen evaluation: one prediction per volume
100 B-scans
frozen
per B-scan
B-scan vectors
volume
trained head
$f_{\bar{\theta}}$
mean over
patches
mean over
B-scans
LayerNorm
+ linear
sigmoid:
p(glaucoma)
full view
EMA encoder
256 × 768
100 × 768
768
Secondary variants: patch max per B-scan; max, or mean and max, over B-scans.